

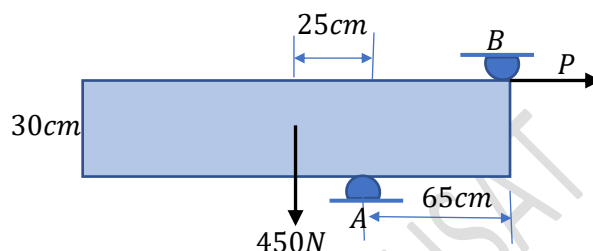
GENERAL CASE OF FORCES IN A PLANE

When the system of forces cannot be exclusively classified as concurrent or parallel, we are dealing with a general case of force system. The equilibrium equations for a general case of force system are

$$\Sigma F_x = 0 \quad \langle \rangle \quad \Sigma F_y = 0 \quad \langle \rangle \quad \Sigma M = 0$$

Example 1

A heavy prismatic timber of weight $W = 450 \text{ N}$ and length $l = 180 \text{ cm}$ is supported horizontally between two fixed fulcrums A and B . If the coefficient of friction between the timber and each fulcrum is $\mu = \frac{1}{3}$, find the magnitude of the horizontal force P applied as shown that will cause impending sliding of the timber to the right.



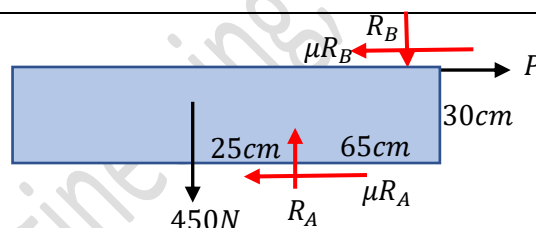
$$\Sigma F_x = 0 \gg P - \mu R_A - \mu R_B = 0 \gg R_A + R_B = 3P$$

$$\Sigma F_y = 0 \gg R_A - R_B - 450 \text{ N} = 0 \gg R_A - R_B = 450$$

$$R_B = \frac{3P - 450}{2}$$

$$\Sigma M_A = 0 \gg$$

$$450 \times 25 + \mu R_B \times 30 - R_B \times 65 - P \times 30 = 0$$



$$P \times 30 = 11250 - 55R_B = 11250 - 55 \left(\frac{3P - 450}{2} \right)$$

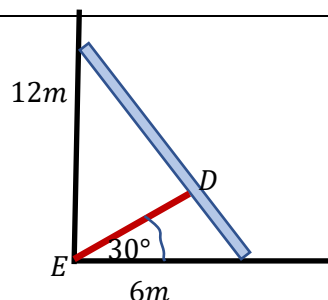
$$112.5P = 23625$$

$$P = 210 \text{ N} \quad \langle \rangle \quad R_A = 540 \text{ N} \quad \langle \rangle \quad R_B = 90 \text{ N}$$

Exercise 14

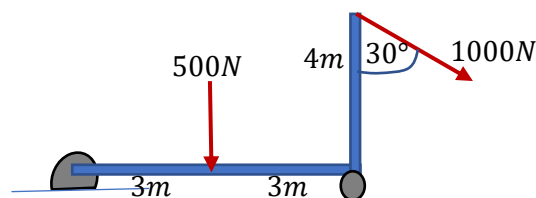
14.1

A 700 N man stands on the middle rung of a 250 N ladder. Assuming that the floor and the wall are perfectly smooth and that slipping is prevented by a string DE , find the tension in the string and the reactions at the supports.



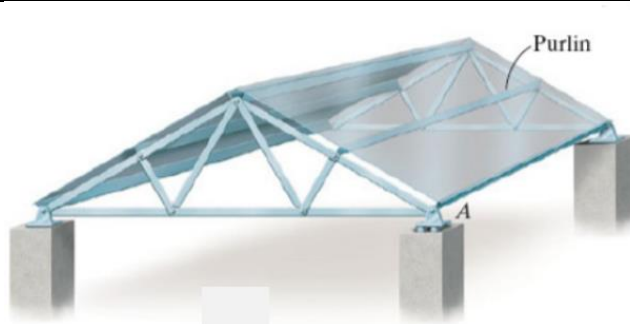
14.2

Calculate the support reactions at the supports.

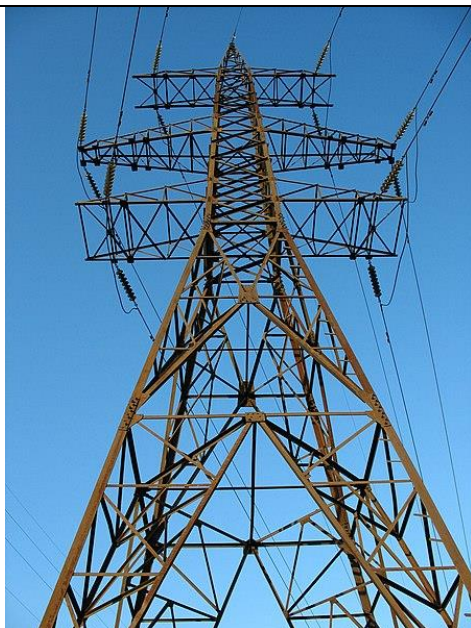


TRUSSES AND FRAMES

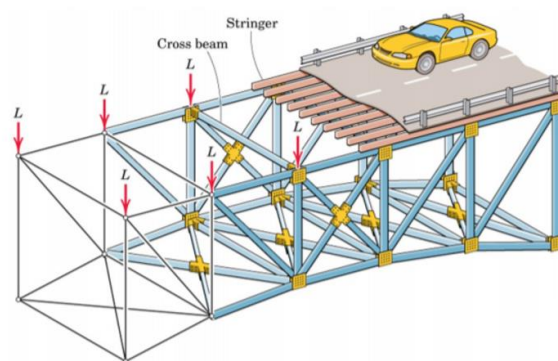
A truss is a rigid structure composed of slender members **joined together at their ends** with the help of bolts, rivets, or welds and **loading is done only at the joints**. If all the members of the truss and all the loads acting on it can be represented on a plane, it is called a plane truss and otherwise a space truss.



Roof truss (planar truss)

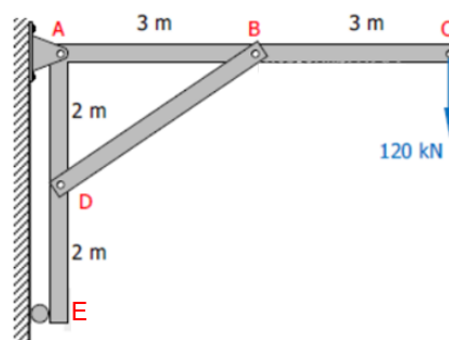


Transmission tower (Space truss)



Bridge truss (Space truss)

The assumption that all the members are joined at their ends and all loads are acting on the joints make each member of the truss as a **two-force** member. **The line of action of the force in each member will then be along the axis of the member.** Hence, all the members in a truss will be either in **tension or compression**. Bending of members is neglected. If these conditions are not satisfied, the structure is called a **frame**. So, in frames we can not assume that the forces in all the members will be along their axes.



frame

If the number of members (n) in a truss is given by $n = 2j - 3$ where j is the number of joints, it is called a **perfect truss**. If $n > 2j - 3$, it is **redundant truss** and if $n < 2j - 3$, it is a **deficient truss**. Perfect trusses are statically determinate structures. Redundant trusses are statically indeterminate structures, and deficient trusses may not remain in equilibrium. Here, our focus is only on perfect truss.

Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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